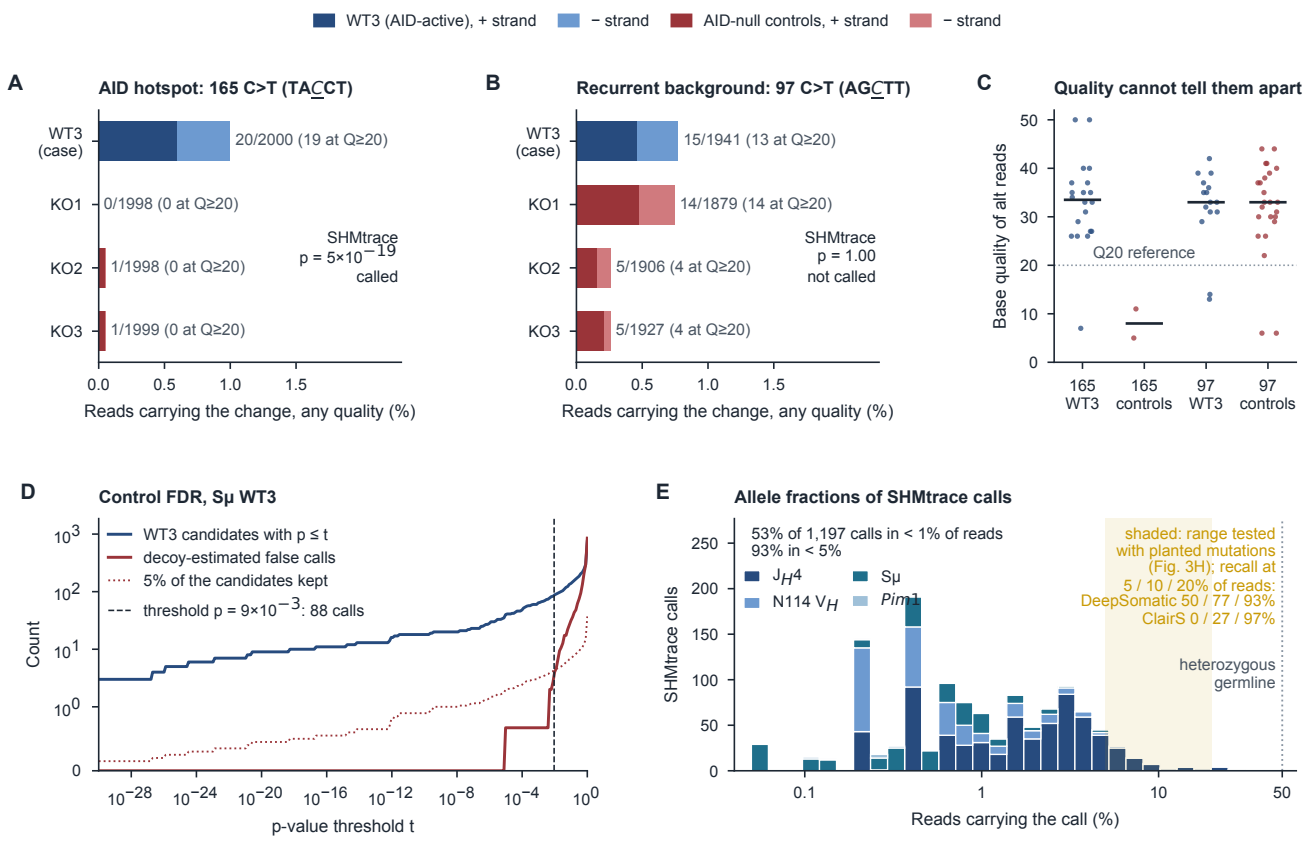

WT3 (AID-active), + strand
− strand
AID-null controls, + strand
− strand
A
AID hotspot: 165 C>T (TACCT)
WT3 (case)
20/2000 (19 at Q≥20)
KO1
0/1998 (0 at Q≥20)
KO2
1/1998 (0 at Q≥20)
KO3
1/1999 (0 at Q≥20)
SHMtrace
p = 5×10−19
called
0.0
0.5
1.0
1.5
Reads carrying the change, any quality (%)
B
Recurrent background: 97 C>T (AGCTT)
WT3 (case)
15/1941 (13 at Q≥20)
KO1
14/1879 (14 at Q≥20)
KO2
5/1906 (4 at Q≥20)
KO3
5/1927 (4 at Q≥20)
SHMtrace
p = 1.00
not called
0.0
0.5
1.0
1.5
Reads carrying the change, any quality (%)
C
Quality cannot tell them apart
Base quality of alt reads
50
40
30
20
10
0
Q20 reference
165 WT3
165 controls
97 WT3
97 controls
D
Control FDR, Sμ WT3
WT3 candidates with p ≤ t
decoy-estimated false calls
5% of the candidates kept
threshold p = 9×10−3: 88 calls
Count
10^3
10^2
10^1
10^0
0
10−28
10−24
10−20
10−16
10−12
10−8
10−4
100
p-value threshold t
E
Allele fractions of SHMtrace calls
53% of 1,197 calls in < 1% of reads
93% in < 5%
JH4
N114 VH
Sμ
Pim1
shaded: range tested with planted mutations (Fig. 3H); recall at 5 / 10 / 20% of reads:
DeepSomatic 50 / 77 / 93%
ClairS 0 / 27 / 97%
heterozygous germline
SHMtrace calls
250
200
150
100
50
0
0.1
1
10
50
Reads carrying the call (%)